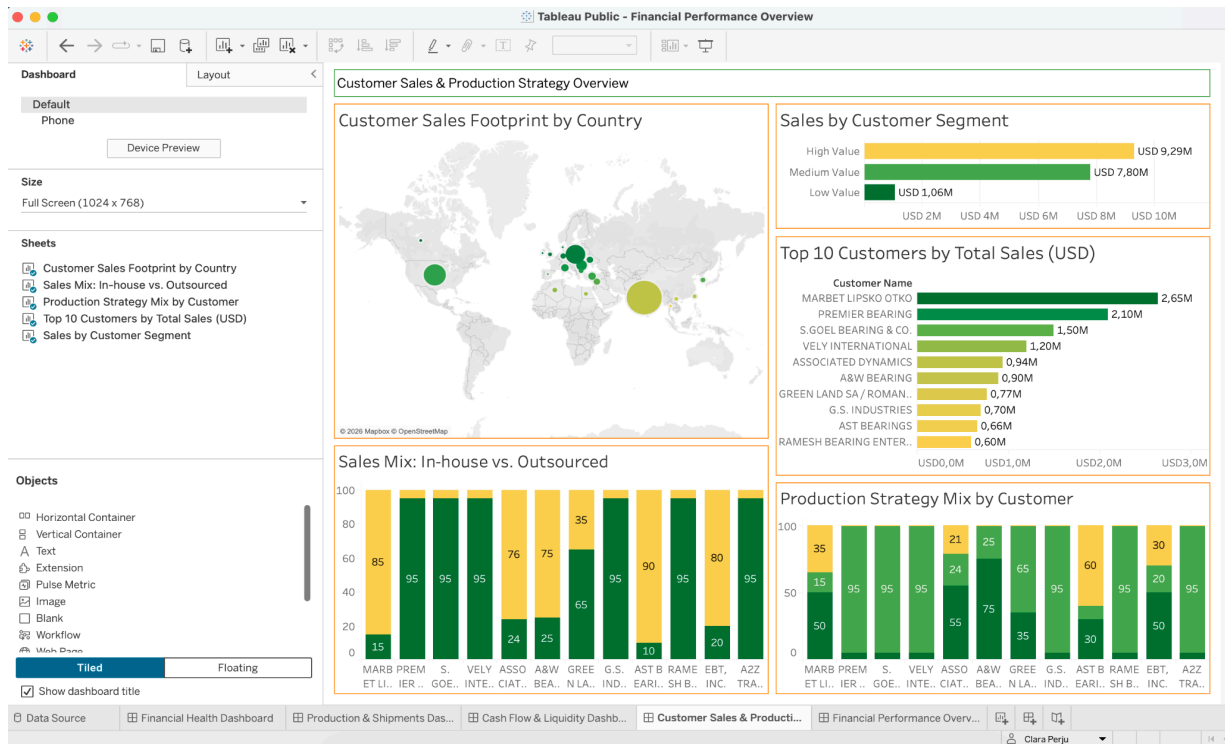


FINANCIAL ANALYTICS PIPELINE

Clara Perju – Financial Analyst

Documentation & Technical Project Overview – Manufacturing Enterprise

- Background: BSc Finance & Banking | MSc Accounting, Control & Audit
- [Tableau Dashboards](#)
- [GitHub](#)



1. Executive Summary

Engineered an end-to-end data engineering pipeline and analytics warehouse utilizing financial and operational data from a real manufacturing company (anonymized for privacy). The architecture transforms raw, multi-sheet financial Excel reports via Python script into a 3-layer MySQL data warehouse (Bronze, Silver, Gold) featuring automated ingestion, SQL-driven transformations, and executive Tableau dashboards.

Note: All financial and operational metrics originate from a real manufacturing enterprise. To comply with non-disclosure and privacy agreements, company identifiers have been completely anonymised, and public links serve strictly as demonstrative references.

2. Medallion Architecture & Data Flow

The architecture cleanses and refines data across three distinct tiers:

- **Bronze Layer** (Raw Data Ingestion – 12 tables): Automated Python scripts read multi-sheet Excel files. Data wide-to-long reshaping is handled via `pandas.melt()` and `pivot_table()`, converting unformatted layouts into structured rows and handling missing values (NaN to None) for clean database insertion.
- **Silver Layer** (Normalized Schema – 14 tables): Data is stored in a normalized relational schema with primary and foreign keys. Cleaning includes regex standardization (`REGEXP_REPLACE`), dynamic date parsing (`CAST + CASE WHEN`), and type casting (e.g., `FLOAT` to `DECIMAL(15,2)`).
- **Gold Layer** (Analytics & BI Views – 5 views): High-performance analytical views with SQL-calculated business KPIs, structured for immediate consumption by Tableau dashboards.

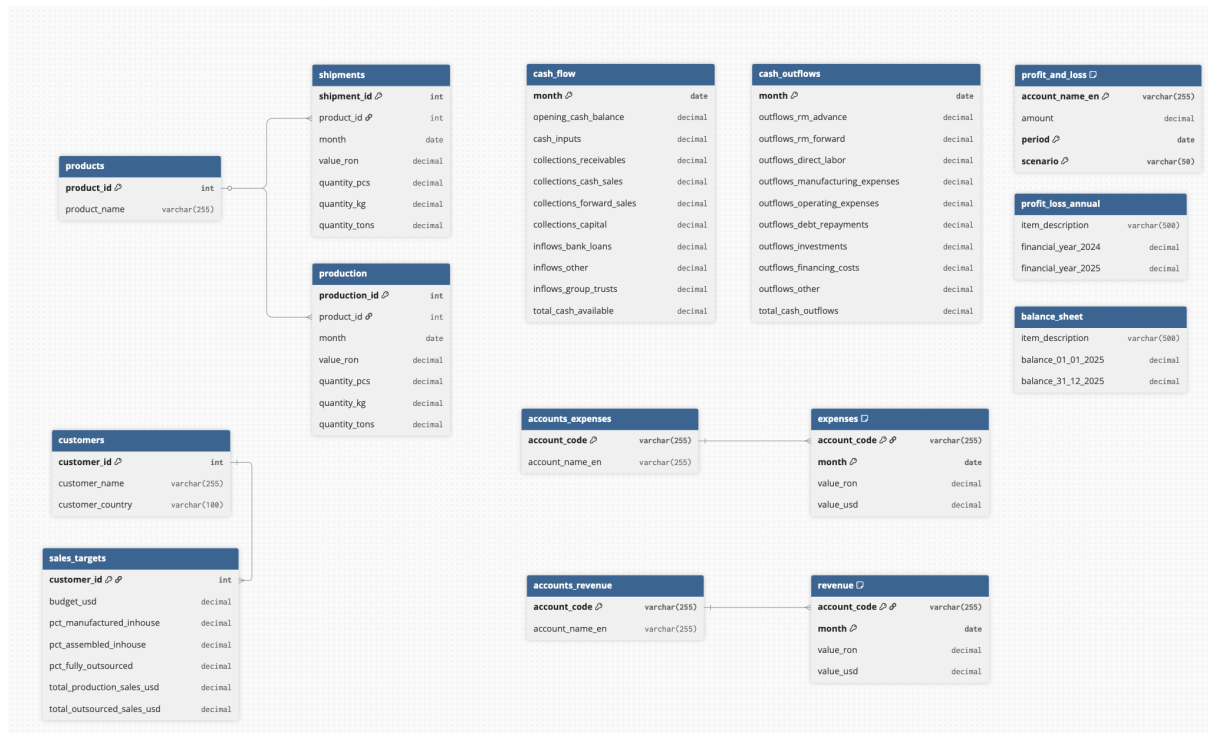
3. Key Technical Capabilities

- Python Data Engineering: Programmatic extraction of multi-sheet workbooks using `pandas`, `mysql-connector-python`, and `python-dotenv`.
- Advanced SQL Transformations: Use of window functions (`LAG()`, `SUM() OVER()`) for YTD trends, dynamic type casting, and defensive coding via `NULLIF` to prevent zero-division errors in financial metrics.
- Financial KPI Engine: SQL-driven computation of over 20 core ratios covering profitability, liquidity, operational efficiency, and leverage.
- Interactive Dashboards: 5 Tableau dashboards covering operations, cash flow, P&L, and customer analytics with conditional formatting and drill-down details.

4. Tech Stack & Data Sources

Tech Stack:

- Data Ingestion: Python 3.9 (`pandas`, `mysql-connector-python`, `python-dotenv`) – ETL automation
- Data Storage: MySQL 8.0 – 3-layer data warehouse
- Data Transformation: SQL – Cleaning, typing, KPI calculation
- Data Visualization: Tableau Public – Executive dashboards
- [Data Modeling](#)



5. Gold Layer – Analytical Views (5 Views)

View	Analytics Value
vw_production_vs_shipments	Fill rate %, inventory gap, MoM production growth, YTD totals operational performance tracking.
vw_cash_overview	Net cash, MoM change, cash flow ratio, outflow % liquidity monitoring.
vw_revenue_expenses_overview	Net profit, margins, MoM changes, cumulative trends P&L performance.
vw_sales_customer_overview	Budget vs actuals, in-house vs outsourced split customer portfolio analysis.
vw_financial_ratios	ROE, ROA, EBITDA, current ratio, debt-to-equity executive financial health scorecard.

7. Key KPIs Calculated

- Profitability: Gross Margin %, Operating Margin %, Net Margin %, ROE, ROA, EBITDA & EBITDA Margin %
- Liquidity: Current Ratio, Quick Ratio, Cash Ratio, Working Capital, Net Debt
- Leverage: Debt to Equity, Debt to Assets, Long-term Debt to Equity, Equity Ratio %
- Operations: Fill Rate %, Inventory Gap, MoM Revenue & Expense Change %, YTD Production & Shipments

8. What I Learned

- Designing and implementing a Medallion architecture from scratch on real industrial dataset structures.
- Data modeling: normalization, composite keys, foreign keys, natural vs. surrogate key trade-offs.
- Python/pandas wide-to-long format transformation using `melt()`, `pivot_table()`, `ffill()`, `transpose()`.
- Advanced SQL: CASE WHEN, COALESCE, REGEXP_REPLACE, STR_TO_DATE, window functions (SUM OVER, LAG, PARTITION BY), CROSS JOIN for ratio analysis.
- Financial analytics: calculating and interpreting profitability, liquidity, and leverage ratios from balance sheet and P&L data.
- Tableau: dual-axis charts, calculated fields, reference lines, conditional formatting.